THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012: Fundamental Concepts of Mathematics

Dec 8, 2023 2:30 p.m. – 5:00 p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL 7 questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (8 points) **Prove or disprove** the following statement: For any nonempty sets A, B,

$$\mathcal{P}(A \setminus B) = (\mathcal{P}(A) \setminus \mathcal{P}(B)) \cup \{\emptyset\}.$$

The following solution is generated by an AI-based tool (with slight modification of wordings).

If you think you should prove the given statement, comment and find out the missing gap(s) in the following proof (if any).

If you think you should disprove it, point out the mistake(s) and explain briefly what goes wrong. Then provide a disproof.

> "Proof": To prove that these two sets are equal, we need to show that every element of $\mathcal{P}(A \setminus B)$ is also an element of $(\mathcal{P}(A) \setminus \mathcal{P}(B)) \cup \{\emptyset\}$, and conversely, every element of $(\mathcal{P}(A) \setminus \mathcal{P}(B)) \cup \{\emptyset\}$ is also an element of $\mathcal{P}(A \setminus B)$.
> Every element X of $\mathcal{P}(A \setminus B)$ is a subset of $A \setminus B$. If X is the empty set, then it belongs to $\{\emptyset\}$. If X is nonempty, then it is a subset of A and not a subset of B. Hence X is an element of $\mathcal{P}(A) \setminus \mathcal{P}(B)$.
> Conversely, every element X of $(\mathcal{P}(A) \setminus \mathcal{P}(B)) \cup \{\emptyset\}$ is either in $\mathcal{P}(A) \setminus \mathcal{P}(B)$ or in $\{\emptyset\}$. If X is in $\{\emptyset\}$, then it is the empty set, which is also an element of $\mathcal{P}(A \setminus B)$. If X is in $\mathcal{P}(A) \setminus \mathcal{P}(B)$, then it does not contain any element from B and is therefore a subset of $A \setminus B$. Hence X is an element of $\mathcal{P}(A \setminus B)$.

2. (24 points) **Prove or disprove** each of the following statements:

   (a) The set $A = \{(x, y) \in \mathbb{Q} \times \mathbb{R} : x + y + 1 = 0\}$ is denumerable.

   (b) For any sets A, B, if A, B are uncountable, then $A \triangle B$ is uncountable.

   (c) Let A be a nonempty set and let f be a relation on A which is both a transitive relation and a function from A to A. Then f is bijective.

   (d) The only number in the sequence $a_1 = 4, a_2 = 44, a_3 = 444, a_4 = 4444, \ldots$ which is a perfect square is the first term $a_1 = 4$.

3. (18 points)

(a) Prove that if there exist nonnegative integers k, l, m, n satisfying the equation

$$4^k + 5^l + 6^m = 7^n, \qquad (*)$$

then l must be odd and $m = 0$. (*Hint*: Consider modulo 3.)

(b) Hence, or otherwise, prove that $k = 0$.

(c) Use the results of (a) and (b) to prove that $(k, l, m, n) = (0, 1, 0, 1)$ is the only nonnegative integer solution to $(*)$.

4. (14 points) For any sets A, B, C, D, consider the following system of set equalities:

$$\begin{cases} A \cap X = B, \\ C \cap X = D. \end{cases}$$

Give a necessary and sufficient condition on the sets A, B, C, D for the existence of a set X satisfying the above system. Provide a detailed proof to justify your claim.

5. (10 points) Let $f : \mathbb{N} \longrightarrow \mathbb{Z}$ be a function satisfying

(i) $f(2) = 2$,

(ii) for any $m, n \in \mathbb{N}$, $f(mn) = f(m)f(n)$, and

(iii) for any $m, n \in \mathbb{N}$, if $m > n$, then $f(m) > f(n)$.

Use induction to prove that $f(n) = n$ for any $n \in \mathbb{N}$.

6. (10 points) For any nonempty sets A, B, define the *projection from $A \times B$ to A* to be the function $\pi_A : A \times B \longrightarrow A$ given by $\pi_A(a, b) = a$ for any $(a, b) \in A \times B$. Similarly, define the *projection from $A \times B$ to B* to be the function $\pi_B : A \times B \longrightarrow B$ given by $\pi_B(a, b) = b$ for any $(a, b) \in A \times B$.

Prove or disprove each of the following statements:

(a) For any subset $X \subseteq A$ and for any subset $Y \subseteq B$,

$$X \times Y = \pi_A^{-1}(X) \cap \pi_B^{-1}(Y).$$

(b) For any subset $U \subseteq A \times B$,

$$U = \pi_A(U) \times \pi_B(U).$$

7. (16 points) Let A and B be nonempty sets and let $f : A \longrightarrow B$ be a surjective function. For each $b \in B$, define

$$X_b = f^{-1}(\{b\}),$$

the preimage set of $\{b\}$ under f.

(a) Prove that X_b is nonempty for any $b \in B$.

(b) Let $P = \{X_b : b \in B\}$. Prove that P is a partition of A (We shall say that P is a partition *associated to the function* f).

(c) Let $A = \mathbb{Z}$ and $B = \mathbb{Z}_2 \times \mathbb{Z}_3$, and let $f : \mathbb{Z} \longrightarrow \mathbb{Z}_2 \times \mathbb{Z}_3$ be defined by

$$f(a) = ([3a]_2, [2a+1]_3),$$

where $[a]_n$ denotes the equivalence class with respect to the congruence of integers modulo n.

(i) Prove that f is surjective.

(ii) Prove that $f^{-1}(\{([0]_2, [0]_3)\}) = [4]_6$. Write down (without proof) the preimage sets $f^{-1}(\{([b]_2, [c]_3)\})$ for all other $([b]_2, [c]_3) \in \mathbb{Z}_2 \times \mathbb{Z}_3$.

(iii) Use the above results to construct a partition of $\mathbb{Z}$ associated to the function f.

******* **END OF PAPER** *******